MARKING SCHEME

Model Paper — 05

Mathematics I

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Note for teachers:

- Part A: 2 marks for each question (usually 1 mark for a correct method and 1 mark for the correct answer). Total 50 marks.
- Part B: 10 marks for each question. Total 50 marks. The marks for each step are shown in the right-hand column.
- Award full marks for any correct alternative method. For graphs and values read from a graph or a construction, accept a reasonable range.

Part A — Answers

1. Work-days: 6 × 20 = 120. Workers needed for 8 days = 120 ÷ 8 = 15. Extra workers = 15 − 6.

Final answer: 9 workers

(Marks 2)

2. Draw a line through E parallel to AB. Alternate angles: the part of ∠AED next to EA = x. Interior angles on the same side (EF ∥ CD): the other part = 180° − 70° = 110°. So x + 110° = 150°.

Final answer: x = 40°

(Marks 2)

3. $\frac{1}{2x} + \frac{3}{5x} - \frac{3}{10x} = \frac{5+6-3}{10x}$ (LCM = 10x)

Final answer: $\frac{8}{10x} = \frac{4}{5x}$

(Marks 2)

4. The faces of a right triangular prism are 2 triangles and 3 rectangles (a rectangle may be a square). A trapezium cannot be a face.

Final answer: (iii) a trapezium ✓

(Marks 2)

5. 180° ↔ 36 members, so 1° ↔ 36 ÷ 180 = 0.2 member. Fielding: 40° → 40 × 0.2 (= 36 × 40 ÷ 180).

Final answer: 8 members

(Marks 2)

6. E is on the minor arc, so ∠ADB = 180° − 130° = 50° (opposite angles of cyclic quadrilateral AEBD). DC passes through the centre and C is the mid-point of AB, so DC ⊥ AB and DC bisects ∠ADB. ∠CDB = 50° ÷ 2.

Final answer: ∠CDB = 25°

(Marks 2)

7. $3x^2 - 48 = 3(x^2 - 16) = 3(x^2 - 4^2)$

Final answer: 3(x − 4)(x + 4)

(Marks 2)

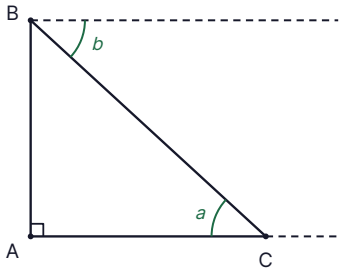
8. $\lg 400 = \lg (4 \times 10^2) = \lg 4 + 2 = 0.6021 + 2$

Final answer: 2.6021

(Marks 2)

9. a = ∠ACB, marked at C between CA and CB. b = angle between the dashed horizontal through B and BC, marked at B. The two horizontal lines are parallel, so a and b are alternate angles.

Final answer: a = b (angle of elevation = angle of depression)



(Marks 2)

10. Curved surface = $2\pi r \times 5r = 10\pi r^2$; base = πr^2. Ratio = $10\pi r^2 : \pi r^2$

Final answer: 10 : 1

(Marks 2)

11. Total outcomes = 4 × 4 = 16. Sum 10: (3, 7), (5, 5), (7, 3) → 3 outcomes.

Final answer: $\frac{3}{16}$

(Marks 2)

12. AB = BD: ∠BDA = ∠DAB = 35°, so exterior ∠DBC = 35° + 35° = 70°. BC = DC: ∠BDC = ∠DBC = 70°. Angles of triangle BCD: x = 180° − 70° − 70°.

Final answer: x = 40°

(Marks 2)

13. A = {2, 3, 5, 7}, B = {3, 6, 9}. A ∩ B = {3} (true). A ∪ B = {2, 3, 5, 6, 7, 9} (so (ii) is false). A′ has 10 − 4 = 6 elements (so (iii) is false).

Final answer: (i) ✓

(Marks 2)

14. $36a^6b^4 \div 6a^3b^2$

Final answer: $6a^3b^2$

(Marks 2)

15. Common ratio $r = (-20) \div (-40) = \frac{1}{2}$. $T_4 = -20 \times \frac{1}{2} = -10$, $T_5 = -10 \times \frac{1}{2}$.

Final answer: −5

(Marks 2)

16. ∠BDC = ∠BAC = 26° (angles in the same segment). BD bisects ∠ADC, so ∠ADC = 52°. ∠CBE = ∠ADC (exterior angle of a cyclic quadrilateral equals the interior opposite angle).

Final answer: ∠CBE = 52°

(Marks 2)

17. HCF of the coefficients 12, 18, 30 = 6; lowest power of a = a; lowest power of b = b^2.

Final answer: $6ab^2$

(Marks 2)

18. Prism volume = 2464 × h. $\pi r^2 h = 2464h \Rightarrow \frac{22}{7} r^2 = 2464 \Rightarrow r^2 = 2464 \times \frac{7}{22} = 784$.

Final answer: r = 28 cm

(Marks 2)

19. DE ∥ BC and BC = 2 × DE = 24 cm (mid-point theorem). $AE^2 = 5^2 + 12^2 = 169$, AE = 13 cm, so EC = AE = 13 cm. DB = AD = 5 cm. Perimeter = DB + BC + CE + ED = 5 + 24 + 13 + 12.

Final answer: 54 cm

(Marks 2)

20. Gradient = $\frac{1-7}{5-2} = -2$. Using (2, 7): $y - 7 = -2(x - 2) \Rightarrow y = -2x + 11$.

Final answer: $y = -2x + 11$ (or $2x + y = 11$)

(Marks 2)

21. ∠DAB = ∠BCD = 70° (opposite angles of a parallelogram). ∠DEB = ∠DAB = 70° (angles in the same segment; A and E are on the same side of DB).

Final answer: ∠DEB = 70°

(Marks 2)

22. Sum = P: $P \times 1.04^2 = 67\,600 \Rightarrow P = 67\,600 \div 1.0816$. (Check: 4% of 1.04P = 2 600 ⇒ 1.04P = 65 000 ⇒ P = 62 500.)

Final answer: Rs. 62 500

(Marks 2)

23. OA ∥ BD: ∠OBD = ∠AOB = 70° (alternate angles). OB = OD, so ∠ODB = 70° and ∠BOD = 180° − 140° = 40°. ∠BED = ½ ∠BOD (angle at centre = 2 × angle at circumference).

Final answer: ∠BED = 20°

(Marks 2)

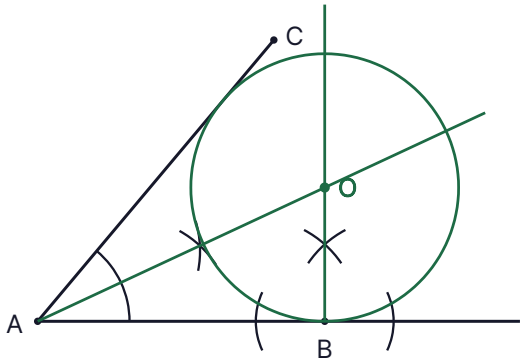
24. $4x > 3 - 15 \Rightarrow 4x > -12 \Rightarrow x > -3$. Negative integers greater than −3:

Final answer: x = −2, −1

(Marks 2)

25. Bisect ∠BAC: the bisector through A passes through the crossing point of the two arcs. At B draw the perpendicular to AB through the crossing point of the two arcs. The bisector and the perpendicular meet at O (OB = distance from O to AC). In the figure O lies directly above B, at a height of AB × tan(25°) ≈ 0.47 AB.

Final answer: O = intersection of the bisector of ∠BAC and the perpendicular to AB at B (1 mark for each line)



(Marks 2)

Part B — Answers and mark allocation

Question 1

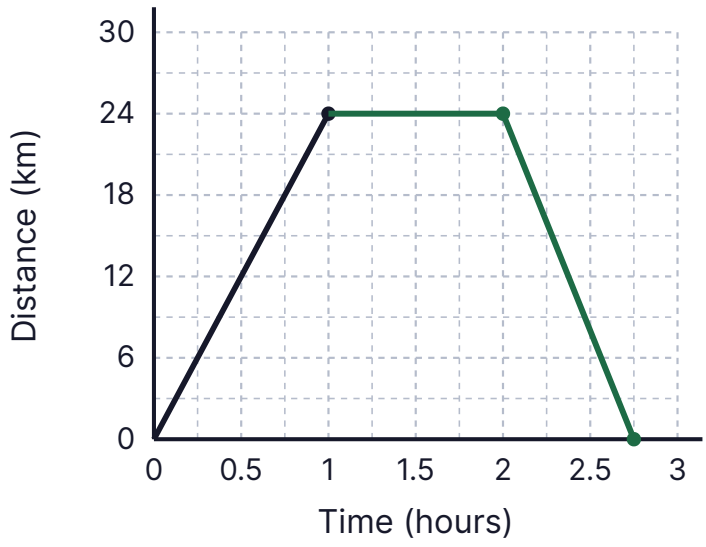
(i)	
$1 - \frac{1}{5} = \frac{4}{5}$	**1**
(ii)	
Other expenses = $\frac{3}{8} \times \frac{4}{5} = \frac{3}{10}$	**1**
Fraction left = $\frac{4}{5} - \frac{3}{10} = \frac{1}{2}$	**1**
(iii)	
Money received = 47 500 ÷ $\frac{1}{2}$ = Rs. 95 000 (or 47 500 × 2)	**1**
Fuel = $\frac{1}{5}$ × 95 000 = Rs. 19 000; other expenses = $\frac{3}{10}$ × 95 000 = Rs. 28 500	**1**
(iv)	
Part of the money received above Rs. 75 000 = 95 000 − 75 000 = Rs. 20 000	**1**
This is 80% of the income above Rs. 75 000 (tax is 20%): income above Rs. 75 000 = 20 000 × 100 ÷ 80 = Rs. 25 000	**1**
Income before tax = 75 000 + 25 000 = Rs. 100 000	**1**
(v)	
Fuel as a percentage of income = 19 000 ÷ 100 000 × 100	**1**
= 19%	**1**

Total marks: 10

Question 2

(a)(i)	
Horizontal line from (1, 24) to (2, 24) (stays for 1 hour at 24 km)	**1**
Straight line from (2, 24) down to (2.75, 0), i.e. 2 h 45 min (accept 2¾ ± one small square)	**1**
(a)(ii)	
Speed = distance ÷ time = $24 \div \frac{3}{4}$	**1**
= 32 km/h	**1**
(b)(i)	
Length of the semicircular arc = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 21 = 66$ m	**1**
Perimeter = 66 + 33 + 42 + 33 (arc + BC + CD + DA)	**1**
= 174 m	**1**
(b)(ii)	
Area of rectangle = $42 \times 33 = 1386$ m^2	**1**
Area of semicircle = $\frac{1}{2} \times \frac{22}{7} \times 21 \times 21 = 693$ m^2	**1**
$1386 \div 693 = 2$ (2 times)	**1**

Total marks: 10



Completed graph (the added parts are shown in green)

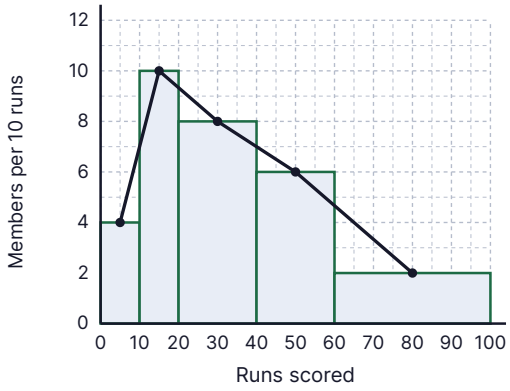
Question 3

(i)	
Rates = 120 000 × $\frac{15}{100}$	**1**
= Rs. 18 000	**1**
(ii)	
Rent for the year = 15 000 × 12 = Rs. 180 000; maintenance = 10% of 180 000 = Rs. 18 000	**1**
Money left = 180 000 − 18 000 − 18 000 = Rs. 144 000	**1**
(iii)(a)	
Number of shares = 12 000 ÷ 6 (or 12 000 × 1/6)	**1**
= 2 000 shares	**1**
Total invested = 2 000 × 80 = Rs. 160 000	**1**
(iii)(b)	
Extra money = 160 000 − 144 000	**1**
= Rs. 16 000	**1**
Percentage = $\frac{16\,000}{160\,000}$ × 100 = 10%	**1**

Total marks: 10

Question 4

Runs scored	Number of members	Cumulative frequency
0 – 10	4	4
10 – 20	10	14
20 – 40	16	30
40 – 60	12	42
60 – 100	8	50



Completed histogram (bars in green)

(i)	
Frequency = height × class width ÷ 10 (height is the number of members per 10 runs)	**1**
10 and 16 (read heights 10, 8 → 10 × 1, 8 × 2)	**1**
12 and 8 (heights 6, 2 → 6 × 2, 2 × 4)	**1**
(ii)	
Bars on the correct class intervals 0–10, 10–20, 20–40, 40–60, 60–100 (widths 10, 10, 20, 20, 40 runs)	**1**
Heights 4, 10, 8, 6, 2 (the tops of the bars lie at the same heights as the plotted points, at the class mid-values)	**1**
Bars adjacent with no gaps; drawn neatly	**1**
(iii)	
Cumulative frequencies 4, 14, 30, 42, 50 (all correct: 2 marks; only 2–3 of the missing values correct: 1 mark)	**2**
(iv)	
Members who scored 40 or more = 12 + 8 = 20 (or 50 − 30)	**1**
Probability = $\frac{20}{50} = \frac{2}{5}$	**1**

Total marks: 10

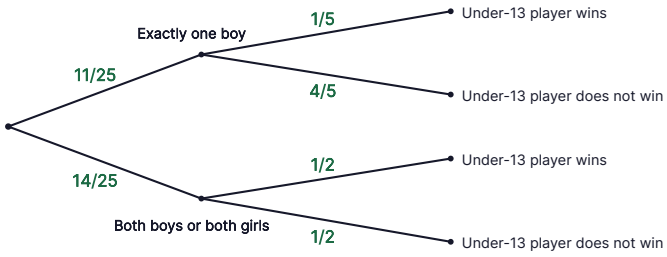
Question 5

(i)	
All 25 outcomes marked with X (all 25: 2 marks; 20 to 24 correct: 1 mark)	**2**
(ii)	
Closed curve(s) enclosing exactly the 11 outcomes with one boy: (G1, G2) with (B4, B5, B6, B7) = 8 and (B1, B2, B3) with G3 = 3 (one loop or two loops accepted)	**1**
Number of favourable outcomes = 11 out of 25	**1**
P(exactly one boy) = $\frac{11}{25}$	**1**
(iii)	
First branches: $\frac{11}{25}$ (exactly one boy) and $\frac{14}{25}$ (both boys or both girls: 12 + 2 = 14)	**1**
Branches after “exactly one boy”: $\frac{1}{5}$ (wins) and $\frac{4}{5}$ (does not win)	**1**
Branches after “both boys or both girls”: $\frac{1}{2}$ and $\frac{1}{2}$	**1**
(iv)	
P(does not win) = $\frac{11}{25} \times \frac{4}{5} + \frac{14}{25} \times \frac{1}{2} = \frac{44}{125} + \frac{7}{25}$	**1**
= $\frac{44}{125} + \frac{35}{125} = \frac{79}{125}$	**1**

Total marks: 10



Sample space and the closed curves



Completed tree diagram